

Principal Component Analysis (PCA)

Dimensionality Reduction in Machine Learning

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Lecture Notes

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1 Introduction to PCA

Principal Component Analysis (PCA) is a foundational technique in Machine Learning used for **dimensionality reduction**.

Before building a predictive model, it is crucial to filter the dataset:

- **Identify Relevant Features:** Keep variables that contribute significant information.
- **Remove Irrelevant/Redundant Features:** Discard noise or correlated variables to simplify the model.

PCA achieves this by transforming the data into a new coordinate system where the new variables (Principal Components) are orthogonal and ordered by the amount of variance they explain.

2 The PCA Algorithm: Step-by-Step

2.1 Step 1: Define the Dataset

Let us consider a dataset with:

- N = Number of examples (data points).
- n = Number of features (variables).

The dataset is represented as a matrix X of size $N \times n$:

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \dots & x_{Nn} \end{bmatrix}$$

Where x_{ik} represents the value of the k -th feature for the i -th example.

2.2 Step 2: Compute Means

Calculate the mean vector $\bar{X}$ containing the mean of each feature column.

$$\bar{x}_i = \frac{1}{N} \sum_{k=1}^N x_{ki} \tag{1}$$

This allows us to understand the central tendency of each dimension.

2.3 Step 3: Compute the Covariance Matrix

The Covariance Matrix (S) captures how features vary with respect to each other. Since there are n features, S will be an $n \times n$ matrix.

[Covariance Formula] For any two features i and j , the covariance is calculated as:

$$\text{Cov}(i, j) = \frac{1}{N-1} \sum_{k=1}^N (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

The resulting matrix S is symmetric:

$$S = \begin{bmatrix} \text{Var}(1) & \text{Cov}(1, 2) & \dots & \text{Cov}(1, n) \\ \text{Cov}(2, 1) & \text{Var}(2) & \dots & \text{Cov}(2, n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(n, 1) & \text{Cov}(n, 2) & \dots & \text{Var}(n) \end{bmatrix}$$

2.4 Step 4: Calculate Eigenvalues and Eigenvectors

To identify the principal components, we solve for the eigenvalues (λ) and eigenvectors (U) of the covariance matrix S .

1. **Characteristic Equation:** Solve determinant $|S - \lambda I| = 0$ to find n roots $(\lambda_1, \lambda_2, \dots, \lambda_n)$.
2. **Eigenvector Equation:** For each λ , solve $(S - \lambda I)U = 0$ to find the corresponding vector.

2.5 Step 5: Normalize Eigenvectors

The raw eigenvectors may have arbitrary scaling. We standardize them to have a length of 1 (Unit Vectors).

If an eigenvector is $V = [v_1, v_2, \dots, v_n]^T$, its magnitude is $\|V\| = \sqrt{v_1^2 + \dots + v_n^2}$.

$$E_{\text{normalized}} = \frac{V}{\|V\|} \quad (2)$$

Repeat this for all n eigenvectors to obtain a set of basis vectors $\{e_1, e_2, \dots, e_n\}$.

2.6 Step 6: Order and Select Components

The eigenvalues represent the amount of variance captured by each eigenvector.

- **Sort:** Order eigenvalues from highest to lowest: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.
- **Select:** Choose the top p components based on the desired dimensionality (where $1 \leq p \leq n$).

Selection Logic

If we want to reduce 10 features ($n = 10$) to 5 features, we select the eigenvectors corresponding to the top 5 largest eigenvalues ($p = 5$).

2.7 Step 7: Derive New Dataset (Projection)

Construct a projection matrix W using the selected p normalized eigenvectors.

$$W_{n \times p} = [e_1, e_2, \dots, e_p]$$

Transform the original data X (after mean centering) into the new subspace Y :

$$Y_{N \times p} = (X - \bar{X}) \times W \quad (3)$$

The result is a new dataset with N examples and only p features, containing the most significant information from the original set.

3 Summary

1. **Standardize:** Compute means (and often scale variance).
2. **Covariance:** Compute matrix S to measure relationships.
3. **Decompose:** Find eigenvectors (direction) and eigenvalues (magnitude).
4. **Sort & Slice:** Keep only the vectors with the highest eigenvalues.
5. **Project:** Map original data onto these new vectors.

4 Practice Problems

Problem 1: Matrix Dimensions

You have a dataset with 500 examples and 20 features. You perform PCA and select the top 3 components.

1. What is the size of the Covariance Matrix?
2. What is the size of the final reduced dataset?

Problem 2: Eigenvalue Significance

Given eigenvalues $\lambda_1 = 4.5, \lambda_2 = 0.8, \lambda_3 = 2.1$.

1. Which eigenvalue corresponds to the first Principal Component?
2. If you keep only the first PC, roughly what percentage of variance is lost? (Assume total variance = sum of λ).

Problem 3: Normalization

An eigenvector is calculated as $v = [3, 4]^T$. Calculate the normalized unit vector e .

5 Introduction

Principal Component Analysis (PCA) is a widely used statistical technique in Machine Learning, primarily employed for **dimensionality reduction**. Its main objective is to transform a high-dimensional dataset into a lower-dimensional form while preserving the maximum amount of information (variance).

Key Concepts

- **High Dimensionality:** A dataset with many features (variables).
- **Goal:** Reduce n dimensions to k dimensions (where $k < n$).
- **Variance:** A measure of the spread of data; PCA aims to maximize this in the new dimensions.

6 Problem Statement

Consider a dataset with **4 examples** and **2 features** (X_1 and X_2). The objective is to reduce this data from 2 dimensions to 1 dimension (1D) using PCA.

Example	Feature X_1	Feature X_2
1	4	11
2	8	4
3	13	5
4	7	14

Table 1: Initial 2D Dataset

7 Step-by-Step PCA Calculation

7.1 Step 1: Calculate the Mean of Features

The first step is to calculate the arithmetic mean ($\bar{X}$) for each feature column.

$$\bar{X} = \frac{\sum X_i}{n} \quad \text{where } n = 4 \quad (4)$$

Calculations:

- **Mean of X_1 ($\bar{X}_1$):**

$$\bar{X}_1 = \frac{4 + 8 + 13 + 7}{4} = \frac{32}{4} = 8$$

- **Mean of X_2 ($\bar{X}_2$):**

$$\bar{X}_2 = \frac{11 + 4 + 5 + 14}{4} = \frac{34}{4} = 8.5$$

7.2 Step 2: Calculate the Covariance Matrix

Construct the Covariance Matrix S . Since there are 2 features, S will be a 2×2 matrix.

$$S = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$$

Formula for Sample Covariance:

$$\text{Cov}(A, B) = \frac{1}{n-1} \sum_{k=1}^n (A_k - \bar{A})(B_k - \bar{B}) \quad (5)$$

Calculations:

1. **Cov**(X_1, X_1) (Variance of X_1):

$$\begin{aligned} &= \frac{1}{3}[(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2] \\ &= \frac{1}{3}[(-4)^2 + 0^2 + 5^2 + (-1)^2] \\ &= \frac{1}{3}[16 + 0 + 25 + 1] = \frac{42}{3} = \mathbf{14} \end{aligned}$$

2. **Cov**(X_1, X_2):

$$\begin{aligned} &= \frac{1}{3}[(4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5)] \\ &= \frac{1}{3}[(-4)(2.5) + (0)(-4.5) + (5)(-3.5) + (-1)(5.5)] \\ &= \frac{1}{3}[-10 + 0 - 17.5 - 5.5] = \frac{-33}{3} = \mathbf{-11} \end{aligned}$$

3. **Cov**(X_2, X_1): Symmetric to **Cov**(X_1, X_2), so $\mathbf{-11}$.

4. **Cov**(X_2, X_2) (Variance of X_2):

$$\begin{aligned} &= \frac{1}{3}[(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2] \\ &= \frac{1}{3}[6.25 + 20.25 + 12.25 + 30.25] = \frac{69}{3} = \mathbf{23} \end{aligned}$$

Final Covariance Matrix

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

7.3 Step 3: Calculate Eigenvalues

Find the eigenvalues (λ) by solving the characteristic equation: $|S - \lambda I| = 0$.

$$\det \begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix} = 0$$

Expand the determinant:

$$\begin{aligned} (14 - \lambda)(23 - \lambda) - (-11)(-11) &= 0 \\ (322 - 14\lambda - 23\lambda + \lambda^2) - 121 &= 0 \\ \lambda^2 - 37\lambda + 201 &= 0 \end{aligned}$$

Solving the quadratic equation yields:

$$\lambda_1 \approx \mathbf{30.385} \quad (\text{Largest Eigenvalue})$$

$$\lambda_2 \approx \mathbf{6.615}$$

7.4 Step 4: Calculate Eigenvectors

To reduce dimensions, select the **largest eigenvalue** ($\lambda_1 = 30.385$) as it captures the maximum variance. Solve for Eigenvector U :

$$[S - \lambda_1 I]U = 0$$

$$\begin{bmatrix} 14 - 30.385 & -11 \\ -11 & 23 - 30.385 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -16.385 & -11 \\ -11 & -7.385 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From row 1: $-16.385U_1 - 11U_2 = 0 \implies U_1 \approx 11, \quad U_2 \approx -16.385$.

Normalize to get Unit Eigenvector (e_1):

$$\|U\| = \sqrt{11^2 + (-16.385)^2} \approx 19.73$$

$$e_1 = \begin{bmatrix} 11/19.73 \\ -16.385/19.73 \end{bmatrix} \approx \begin{bmatrix} \mathbf{0.5574} \\ \mathbf{-0.8303} \end{bmatrix}$$

7.5 Step 5: Derive Principal Components

Transform original 2D data into 1D using:

$$Y = e_1^T \times (X - \bar{X}) \tag{6}$$

Example Calculation for Data Point 1 (4, 11):

$$Y_1 = [0.5574 \quad -0.8303] \times \begin{bmatrix} 4 - 8 \\ 11 - 8.5 \end{bmatrix}$$

$$Y_1 = [0.5574 \quad -0.8303] \times \begin{bmatrix} -4 \\ 2.5 \end{bmatrix}$$

$$Y_1 \approx (0.5574)(-4) + (-0.8303)(2.5)$$

$$Y_1 \approx -2.2296 - 2.07575 \approx \mathbf{-4.305}$$

Example	X_1	X_2	New Feature (PC1)
1	4	11	-4.305
2	8	4	3.736
3	13	5	5.693
4	7	14	-5.123

Table 2: Final Transformed Dataset

8 Geometric Interpretation

- **Plotting:** Original points are plotted in 2D space.
- **Centering:** The origin moves to the mean $(\bar{X}_1, \bar{X}_2)$.
- **Projection:** The eigenvector defines the line of "best fit" (maximum spread). The new 1D values are the projections of the original points onto this line.

9 Practice Problems

Problem 1: Covariance Matrix

Given the dataset: $(2, 4), (4, 8)$. Calculate the Covariance Matrix.

Hint: First compute means $\bar{X}_1$ and $\bar{X}_2$.

Problem 2: Eigenvalues

Find the eigenvalues for the matrix:

$$S = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

Problem 3: Normalization

If an eigenvector is $U = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, calculate the normalized Unit Eigenvector e .

This example provides a comprehensive, step-by-step solution for applying Principal Component Analysis (PCA) to a dataset. PCA is a linear dimensionality reduction technique used in exploratory data analysis and data preprocessing.

Objective: Reduce a dataset with **2 features** (X, Y) and **10 examples** into a **1-dimensional** dataset while retaining the maximum information.

10 Step-by-Step Solution

10.1 Step 1: Compute Feature Means

First, calculate the mean (μ or $\bar{x}$) for every feature column to determine the center of the data.

- **Given Data:** 10 data points.
- **Sum of X:** $\sum X = 18.1$
- **Sum of Y:** $\sum Y = 19.1$

$$\bar{X} = \frac{18.1}{10} = \mathbf{1.81}, \quad \bar{Y} = \frac{19.1}{10} = \mathbf{1.91} \quad (7)$$

10.2 Step 2: Compute the Covariance Matrix

The covariance matrix C describes the spread and directional relationship of the data. For 2 dimensions, C is a 2×2 matrix.

$$C = \begin{bmatrix} \text{Cov}(X, X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Cov}(Y, Y) \end{bmatrix}$$

Calculation Strategy: Create a table to calculate deviations from the mean ($X - \bar{X}$) and ($Y - \bar{Y}$), square them, and find the product of deviations.

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	x^2	y^2	xy
2.5	2.4	0.69	0.49	0.476	0.240	0.338
0.5	0.7	-1.31	-1.21	1.716	1.464	1.585
...	...	...	...	...	...	...
Sum		0	0	5.549	6.449	5.538

Table 3: Intermediate calculations for Covariance

Apply Formula: $\text{Cov}(A, B) = \frac{\sum(A - \bar{A})(B - \bar{B})}{n-1}$. Here $n = 10$, so divisor is 9.

- $\text{Cov}(X, X) = 5.549/9 = \mathbf{0.6166}$
- $\text{Cov}(Y, Y) = 6.449/9 = \mathbf{0.7166}$
- $\text{Cov}(X, Y) = 5.538/9 = \mathbf{0.6154}$

$$C = \begin{bmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{bmatrix}$$

10.3 Step 3: Calculate Eigenvalues

Solve the characteristic equation $|C - \lambda I| = 0$:

$$\det \begin{bmatrix} 0.6166 - \lambda & 0.6154 \\ 0.6154 & 0.7166 - \lambda \end{bmatrix} = 0$$

Expanding the determinant:

$$(0.6166 - \lambda)(0.7166 - \lambda) - (0.6154)^2 = 0$$

$$\lambda^2 - 1.333\lambda + 0.4418 - 0.3787 = 0$$

$$\lambda^2 - 1.333\lambda + 0.0631 = 0$$

Solving the quadratic equation yields two eigenvalues:

Eigenvalues

$$\lambda_1 \approx 0.049 \quad \lambda_2 \approx 1.284$$

10.4 Step 4: Eigenvectors and Normalization

We select the largest eigenvalue ($\lambda_2 = 1.284$) because it captures the most information. We find its eigenvector V .

$$[C - 1.284I]V = 0$$

$$\begin{bmatrix} -0.6674 & 0.6154 \\ 0.6154 & -0.5674 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From row 1: $-0.6674x + 0.6154y = 0 \implies 0.6674x = 0.6154y$.

Let $y = 1 \implies x \approx 0.922$. Raw Vector $V = [0.922, 1]^T$.

Normalization: Compute length: $L = \sqrt{0.922^2 + 1^2} \approx 1.36$. Divide components by length:

$$\text{Normalized } V_2 = \begin{bmatrix} 0.922/1.36 \\ 1/1.36 \end{bmatrix} = \begin{bmatrix} \mathbf{0.677} \\ \mathbf{0.735} \end{bmatrix}$$

10.5 Step 5: Dimensionality Reduction (Projection)

To convert the original 2D data to 1D, we project the original data matrix D onto the principal component V_2 .

$$\text{New Feature} = D \times V_2$$

Example for the first data point (2.5, 2.4):

$$(2.5 \times 0.677) + (2.4 \times 0.735) \approx 1.69 + 1.76 = \mathbf{3.45}$$

Repeating this for all 10 rows gives the final 1-dimensional dataset.

11 Geometric Interpretation

1. **Centering:** The coordinate system shifts so the origin is at the mean (1.81, 1.91).
2. **Rotation:** The axes rotate to align with the eigenvectors.
3. **Principal Component:** The axis corresponding to λ_2 (the long axis of the data "blob") becomes the new X-axis.
4. **Projection:** Points are "dropped" perpendicularly onto this new axis.

12 Practice Problems

Problem 1: Basics

Given a dataset $X = \{2, 4, 6\}$ and $Y = \{3, 5, 7\}$.

1. Calculate the means $\bar{X}$ and $\bar{Y}$.
2. Calculate the Covariance $\text{Cov}(X, Y)$.

Problem 2: Eigenvector Calculation

You are given the matrix $M = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. One of the eigenvalues is $\lambda = 4$. Find the corresponding normalized eigenvector.

Problem 3: Projection

Original data point $P = [10, 5]$. Selected Principal Component $V = [0.8, 0.6]$. Calculate the transformed 1D value for P .

13 Introduction

This lecture covers a simplified, step-by-step example of Principal Component Analysis (PCA) using a dataset with only **3 examples** and **2 features**. This small scale allows us to verify every calculation manually, providing a clear insight into how PCA reduces dimensions by projecting data onto orthogonal vectors.

14 Problem Statement

We are given a dataset with 3 samples and 2 variables (X and Y).

Example	Feature X	Feature Y
1	2	11
2	3	14
3	7	26

Table 4: Original Dataset

15 Step-by-Step Solution

15.1 Step 1: Compute Means and Center Data

To perform PCA, the data must be centered around the origin (Mean = 0).

1. Calculate Means:

- $\bar{X} = \frac{2+3+7}{3} = \frac{12}{3} = \mathbf{4}$
- $\bar{Y} = \frac{11+14+26}{3} = \frac{51}{3} = \mathbf{17}$

2. Calculate Mean Centered Data (Z): Subtract the means from the original values:
 $Z_X = X - 4$ and $Z_Y = Y - 17$.

Original (X, Y)	Calculation	Centered (x, y)
(2, 11)	(2 - 4, 11 - 17)	(-2 , -6)
(3, 14)	(3 - 4, 14 - 17)	(-1 , -3)
(7, 26)	(7 - 4, 26 - 17)	(3 , 9)

$$Z = \begin{bmatrix} -2 & -6 \\ -1 & -3 \\ 3 & 9 \end{bmatrix}$$

15.2 Step 2: Compute the Covariance Matrix

The Covariance Matrix C is calculated using the formula:

$$C = \frac{1}{n-1}(Z^T Z) \quad (8)$$

Where $n = 3$, so the divisor is 2.

Sums of Products:

- $\sum x^2 = (-2)^2 + (-1)^2 + 3^2 = 4 + 1 + 9 = \mathbf{14}$
- $\sum y^2 = (-6)^2 + (-3)^2 + 9^2 = 36 + 9 + 81 = \mathbf{126}$
- $\sum xy = (-2)(-6) + (-1)(-3) + (3)(9) = 12 + 3 + 27 = \mathbf{42}$

Matrix Calculation:

$$C = \frac{1}{2} \begin{bmatrix} 14 & 42 \\ 42 & 126 \end{bmatrix} = \begin{bmatrix} 7 & 21 \\ 21 & 63 \end{bmatrix}$$

15.3 Step 3: Calculate Eigenvalues

Solve the characteristic equation $|C - \lambda I| = 0$:

$$\det \begin{bmatrix} 7 - \lambda & 21 \\ 21 & 63 - \lambda \end{bmatrix} = 0$$

Expand:

$$\begin{aligned} (7 - \lambda)(63 - \lambda) - (21)(21) &= 0 \\ (441 - 7\lambda - 63\lambda + \lambda^2) - 441 &= 0 \\ \lambda^2 - 70\lambda &= 0 \\ \lambda(\lambda - 70) &= 0 \end{aligned}$$

Resulting Eigenvalues

$$\lambda_1 = 70, \quad \lambda_2 = 0$$

Note: $\lambda_2 = 0$ implies that the original data points lie perfectly on a straight line.

15.4 Step 4: Calculate Eigenvectors

1. For $\lambda_1 = 70$ (First Principal Component):

$$\begin{aligned} [C - 70I]V_1 = 0 &\implies \begin{bmatrix} 7 - 70 & 21 \\ 21 & 63 - 70 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \\ \begin{bmatrix} -63 & 21 \\ 21 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \end{aligned}$$

From row 2: $21x - 7y = 0 \implies 21x = 7y \implies \mathbf{y} = 3\mathbf{x}$.

Let $x = 1$, then $y = 3$. Vector $V_1 = [1, 3]^T$. **Normalization:** Length $L = \sqrt{1^2 + 3^2} = \sqrt{10}$.

$$u_1 = \begin{bmatrix} 1/\sqrt{10} \\ 3/\sqrt{10} \end{bmatrix}$$

2. For $\lambda_2 = 0$ (Second Principal Component):

$$[C - 0I]V_2 = 0 \implies \begin{bmatrix} 7 & 21 \\ 21 & 63 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

From row 1: $7x + 21y = 0 \implies x = -3y$. Let $y = 1$, then $x = -3$. Vector $V_2 = [-3, 1]^T$.

Normalization:

$$u_2 = \begin{bmatrix} -3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$$

15.5 Step 5: Projection (Dimensionality Reduction)

We project the **centered data** (Z) onto the First Principal Component (u_1) to get the new 1D feature space.

$$\text{New Data} = Z \cdot u_1$$

- **Point 1:** $[-2, -6] \cdot [1, 3] \times \frac{1}{\sqrt{10}} = \frac{-2-18}{\sqrt{10}} = \frac{-20}{\sqrt{10}}$
- **Point 2:** $[-1, -3] \cdot [1, 3] \times \frac{1}{\sqrt{10}} = \frac{-1-9}{\sqrt{10}} = \frac{-10}{\sqrt{10}}$
- **Point 3:** $[3, 9] \cdot [1, 3] \times \frac{1}{\sqrt{10}} = \frac{3+27}{\sqrt{10}} = \frac{30}{\sqrt{10}}$

16 Geometric Interpretation

1. **Original Plot:** Points $(2, 11)$, $(3, 14)$, $(7, 26)$ are plotted.
2. **Center Shift:** The origin moves to $(4, 17)$.
3. **Eigenvector Direction:** A line is drawn through the new origin with a slope of 3 (vector $[1, 3]$). This line represents the 1st Principal Component.
4. **Alignment:** Since $\lambda_2 = 0$, all points lie exactly on this line. The PCA successfully captures 100% of the information in just 1 dimension.

17 Practice Problems

Problem 1: Covariance Calculation

Given centered data points $A(-1, -2)$ and $B(1, 2)$. Calculate the Covariance Matrix. (Assume $n = 2$, so divide by $2 - 1 = 1$).

Problem 2: Eigenvalues

If a 2×2 matrix has a determinant of 0 and a trace (sum of diagonal elements) of 10, what are the two eigenvalues?

Problem 3: Projection

Project the point $P(4, 4)$ onto the normalized unit vector $u = [\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]^T$.